

YONGHONG ZHANG

Causal Inference · Reliable AI Agents · Climate Finance

PhD Candidate in Economics & Business, Universidad Autónoma de Madrid · Madrid, Spain

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PROFILE

PhD candidate working at the intersection of causal inference, climate finance, and reliable AI for empirical research. I study whether AI agents genuinely perform execution-critical research tasks and whether their outputs are evaluated against valid criteria. My current work spans agent-benchmark measurement validity, execution-grounded evaluation of causal workflows, evidence-grounded auditing of DID studies, and multi-agent measurement of firm-level ESG perceptions from multilingual news. My empirical research examines the environmental co-benefits of China's emissions-trading pilots using DID and spatial econometrics.

RESEARCH INTERESTS

Causal inference and policy evaluation; reliable AI and agent evaluation; climate finance and carbon markets; firm-level ESG measurement; AI-assisted financial decision-making; evidence-grounded empirical research.

EDUCATION

- PhD in Economics & Business**, Universidad Autónoma de Madrid Feb 2024 – Feb 2028 (expected)
Madrid, Spain. Focus: carbon markets, climate finance, environmental policy evaluation, causal inference, AI-assisted empirical research.
Supervisors: José Ricardo Vaz Antunes Pereira Correia; Isabel María Parra Oller.
- MSc in Corporate & Financial Risk Management**, University of Sussex, United Kingdom Sep 2010 – Feb 2012
Graduate training in risk management, finance, econometrics, and economics.
- BSc in Accounting & Business/Management**, Brunel University of London, United Kingdom 2006 – 2010
2+2 articulation: Renmin University of China, School of International Education, Beijing (2006–2008), then Brunel (2008–2010, degree-conferring).

PUBLICATIONS & WORKING PAPERS

- **CausalVerify: An Execution-Grounded Benchmark for LLM Causal Inference Workflows.** Yonghong Zhang, Ricardo Correia, Isabel M. Parra, and Yong Xie. Benchmarks frontier LLMs on causal-inference workflows, pairing method/design interpretation with execution-grounded coefficient recovery. *Under review. Preprint: arXiv:2609.07944.*
- **Evidence-Grounded Auditing of Identification Assumptions in Climate-Policy Causal Evaluations.** Yonghong Zhang, Yong Xie, Isabel M. Parra, and Ricardo Correia. Introduces ARGUS, a bounded LLM auditing system for DID identification assumptions using an assumption–implication–evidence rubric. *Accepted at ClimateNLP 2026, co-located with EMNLP 2026.*
- **The Double Measurement Confound in Agent Benchmarks: De-Scaffolding, Ground-Truth Scoring, and Reliability Beyond the Mean.** Yonghong Zhang et al. Studies how agent benchmarks can mistakenly measure scaffolds and scorers rather than model capability. *Under review. Preprint: arXiv:2609.09218.*
- **Do Carbon Markets Deliver Environmental Co-Benefits? Evidence from China's ETS Pilots.** Yonghong Zhang, Ricardo Correia, and Isabel M. Parra Oller. Difference-in-differences with province and year fixed effects on a 2004–2022 panel, with spatial-spillover analysis. *Working paper.*

SELECTED RESEARCH & AI PROJECTS

- **GLEAM: Multi-Agent Measurement of Firm-Level ESG Perceptions from Multilingual News.** [Project website] Develops a structured LLM-agent pipeline to construct firm-level environmental, social, and governance perception measures from multilingual news. Ongoing work examines their validity and association with firm-level financial outcomes.
- **ComtradeBench / OpenEnv benchmark.** Deterministic, execution-grounded environment for evaluating autonomous agents under realistic API frictions; basis for the Web Agent Track and OpenEnv Custom Track results below. Demo: huggingface.co/spaces/yonghongzhang/comtrade-env

HONORS, AWARDS & COMPETITIVE ACHIEVEMENTS

- **1st Place, Best Defender, AgentBeats × Lambda Security Arena 2026**, UC Berkeley RDI × Lambda; Team MateFin; 91.4% defender win rate across 21 held-out adversarial runs.
- **1st Place (tie), Web Agent Track, AgentX–AgentBeats (Phase 1)**, UC Berkeley RDI; Google DeepMind-sponsored track; Team MateFin; prize: \$8,500 in Google Cloud credits and \$750 in Lambda credits.
- **2nd Place, OpenEnv Custom Track, AgentX–AgentBeats**, UC Berkeley RDI; Team MateFin; prize: \$3,000 in Hugging Face credits and six months of Hugging Face Pro per team member.
- **Honorable Mention (Accésit), Best Poster Competition, UAM Doctoral Week 2026**; Selected Representative Poster for the Economics and Business Doctoral Programme.
- **Third Prize, 2023 Annual Key Research Project, Rizhao Finance Society, China, awarded April 2024.** Project: Climate Change Risk and Financing Constraints: A Case Study of China.
- **Rising Star of the Year, Shandong Vocational University of Foreign Languages, 2023.**

EXPERIENCE

- PhD Candidate**, Universidad Autónoma de Madrid Feb 2024 – present
Doctoral research on carbon markets, climate-policy evaluation, and environmental co-benefits using panel econometrics and causal-inference designs; reproducible empirical workflows in Stata, R, Python-based evaluation workflows, LaTeX/Overleaf, and GitHub.
- Lecturer in Financial Management**, Shandong Vocational University of Foreign Languages Sep 2022 – Apr 2024
Taught Corporate Finance and Financial Accounting (bilingual English/Chinese); mentored competition teams; served as an academic reference

for students applying to postgraduate programmes in Europe.

Founder & Owner, Bubble-Tea & Coffee Retail Chain, China

May 2014 – 2022

Built and ran an independent multi-store chain over ~8 years, including operations, finance, hiring, suppliers, and local marketing.

English Teacher, Beijing Road Middle School, China

2012 – 2014

Taught English to secondary-school students.

SELECTED PRESENTATIONS

- **Poster, Machine Learning Summer School (MLSS 2026), Tübingen** — presented CausalVerify and ARGUS (poster 7.17).
- **ICFAB 2026, University of Surrey** — presented in the Sustainable Finance track.
- **Doctoral Seminar, Universidad Autónoma de Madrid** — presented the China ETS environmental co-benefits study.

METHODS & TECHNICAL SKILLS

Econometrics: difference-in-differences, panel fixed effects, event studies, placebo tests, spatial spillovers, robustness checks. **Machine learning / AI:** LLM agents, retrieval-augmented generation (RAG), evaluation benchmarks, benchmark validity, evidence grounding, agentic workflows, responsible AI. **Research software & tools:** Python-based research and evaluation workflows, Stata, R, Git/GitHub, LaTeX/Overleaf, Hugging Face Spaces. **Data domains:** climate finance, carbon markets, environmental emissions, Chinese provincial panel data, multilingual news, trade/policy datasets. **Communication:** academic writing, research presentations, poster design, English research communication, Chinese-English translation.

TRAINING & SUMMER SCHOOLS

- **Machine Learning Summer School (MLSS 2026)**, Max Planck Institute for Intelligent Systems & University of Tübingen, Germany, 31 August – 11 September 2026. Selected participant; presented a poster.
- **Agentic AI MOOC, UC Berkeley RDI** (Center for Responsible, Decentralized Intelligence): Legendary Tier (top completion tier), Fall 2025.

LANGUAGES

Chinese (native); English (professional working language); Spanish (B1, in progress).